LLM ENGINEERING

Bonus chapter 2

# Bonus: Long-Context Positional Extension

Bethgelab · Self-paced reading

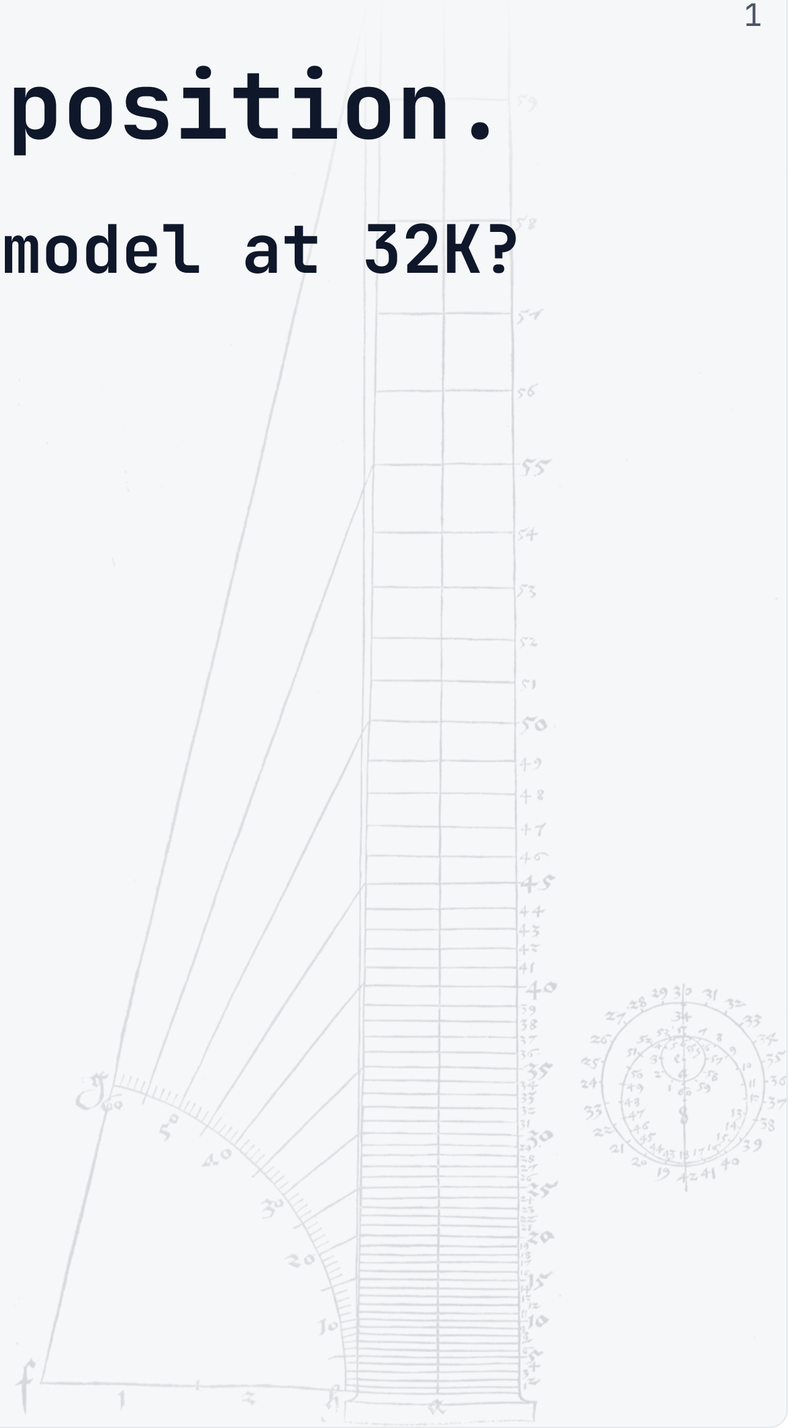
University of Tübingen · Winter term 2026/27

Tübingen AI Center



# RoPE has no maximum position.

## Why not serve an 8K-trained model at 32K?



# Slow pairs leave the trained range

- Illustrative: $b = 10{,}000$ and $d_r = 64$
- Every position still gets an angle
- Slow pairs reach angles never seen in training

| | Within 8K | At 32K |
|---|---|---|
| Slowest pair's largest angle | 1.1 rad | 4.4 rad |
| Pairs below one full turn | 7 of 32 | 2 of 32 |

# Each pair turns at its own rate

(1) $$\omega_j = b^{-2j/d_r}, \qquad \text{angle at position } m = m\,\omega_j, \qquad \Delta = n - m$$

- Extension ratio $\rho = L_t / L_0$: target over trained length
- Fast pairs: short wavelength, many turns
- Slow pairs: long wavelength, few turns
- Every method below reshapes these bands

# PI divides every angle by ρ

(2)
$$m \mapsto \frac{m}{\rho}, \qquad \Delta \mapsto \frac{\Delta}{\rho}, \qquad \omega_j \mapsto \frac{\omega_j}{\rho}$$

- **Position Interpolation** (PI) (Chen et al., 2023)
- Adjacent angle: the rotation one extra token adds
- Position 32,767 maps to 8,191.75
- Fast pairs lose resolution too (Peng et al., 2024)

# Static NTK-aware scaling raises the base

(3)

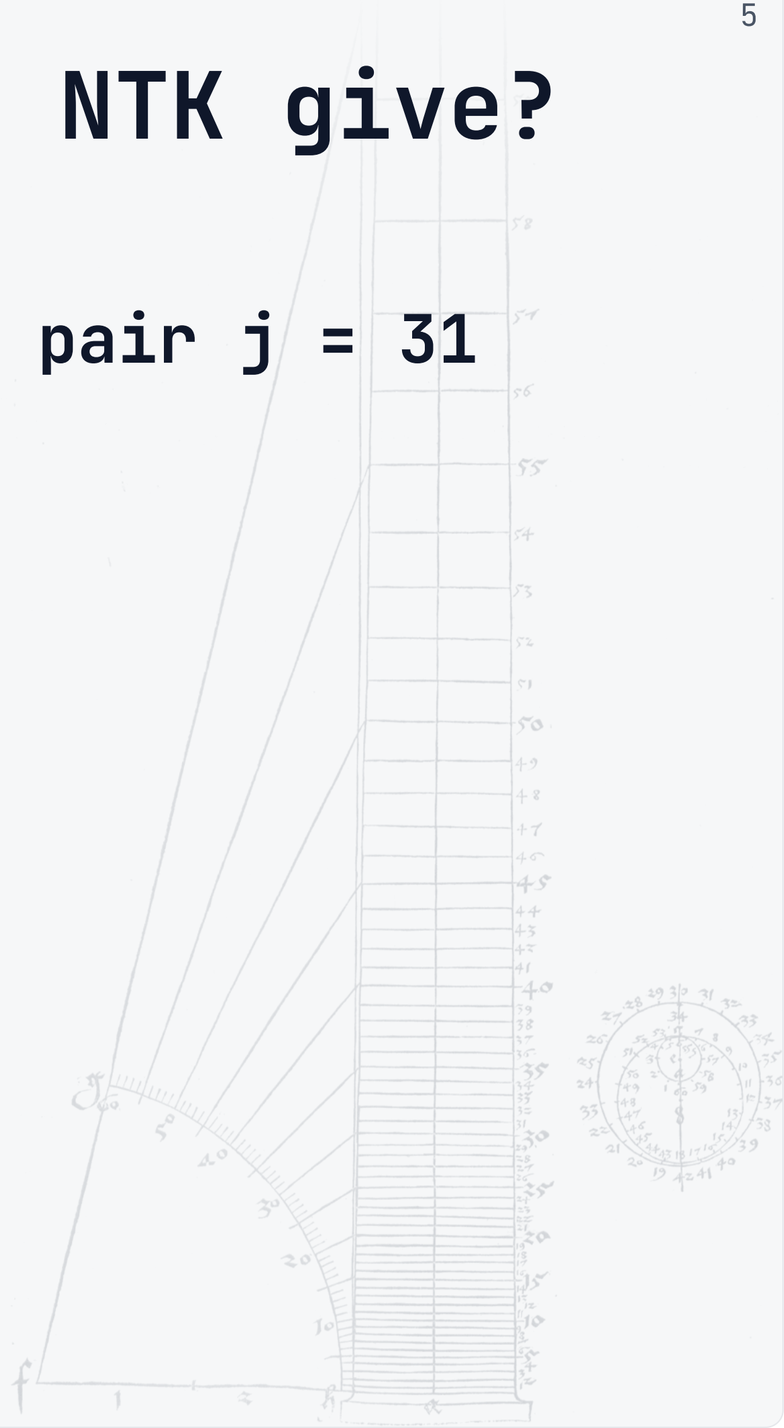
$$b' = b\,\rho^{d_r/(d_r-2)}$$

- Named after the neural tangent kernel (NTK)
- Static: one fixed $\rho$ sets one new base (bloc97, 2023)
- Dynamic NTK: recompute the base as length grows
- Adjusted base frequency (ABF): train with a larger base

# What ratios do PI and NTK give?

$$\rho = 4,\ d_r = 64$$

fastest pair j = 0, slowest pair j = 31



# PI shrinks every pair; NTK spares the fastest

| Adjacent-angle ratio | PI | Static NTK-aware |
|---|---|---|
| Fastest pair, $j = 0$ | 0.25 | 1 |
| Slowest pair, $j = 31$ | 0.25 | 0.25 |

- PI: one factor for every pair
- NTK: the fastest pair keeps its angle

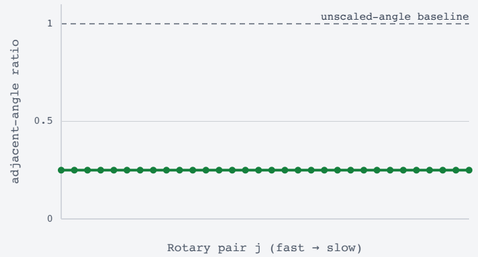
# Compare PI and NTK pair by pair

- Each dot: one rotary pair
- Height: new adjacent angle over the original

**Adjacent-angle ratio**

d_r: 32 | 64 | ρ: 2x | 4x | 8x | method: No extension | Position Interpolation | NTK-aware

L_0=8K, L_t=32K, ρ=4: ratio by rotary pair




Select a rotary-pair dot to read its adjacent-angle ratio.

# YaRN blends by rotation count

(4)
$$\lambda_j = \frac{2\pi}{\omega_j} = 2\pi b^{2j/d_r}, \qquad r_j = \frac{L_0}{\lambda_j}$$

(5)
$$\omega'_j = (1-\gamma_j)\frac{\omega_j}{\rho} + \gamma_j\,\omega_j$$

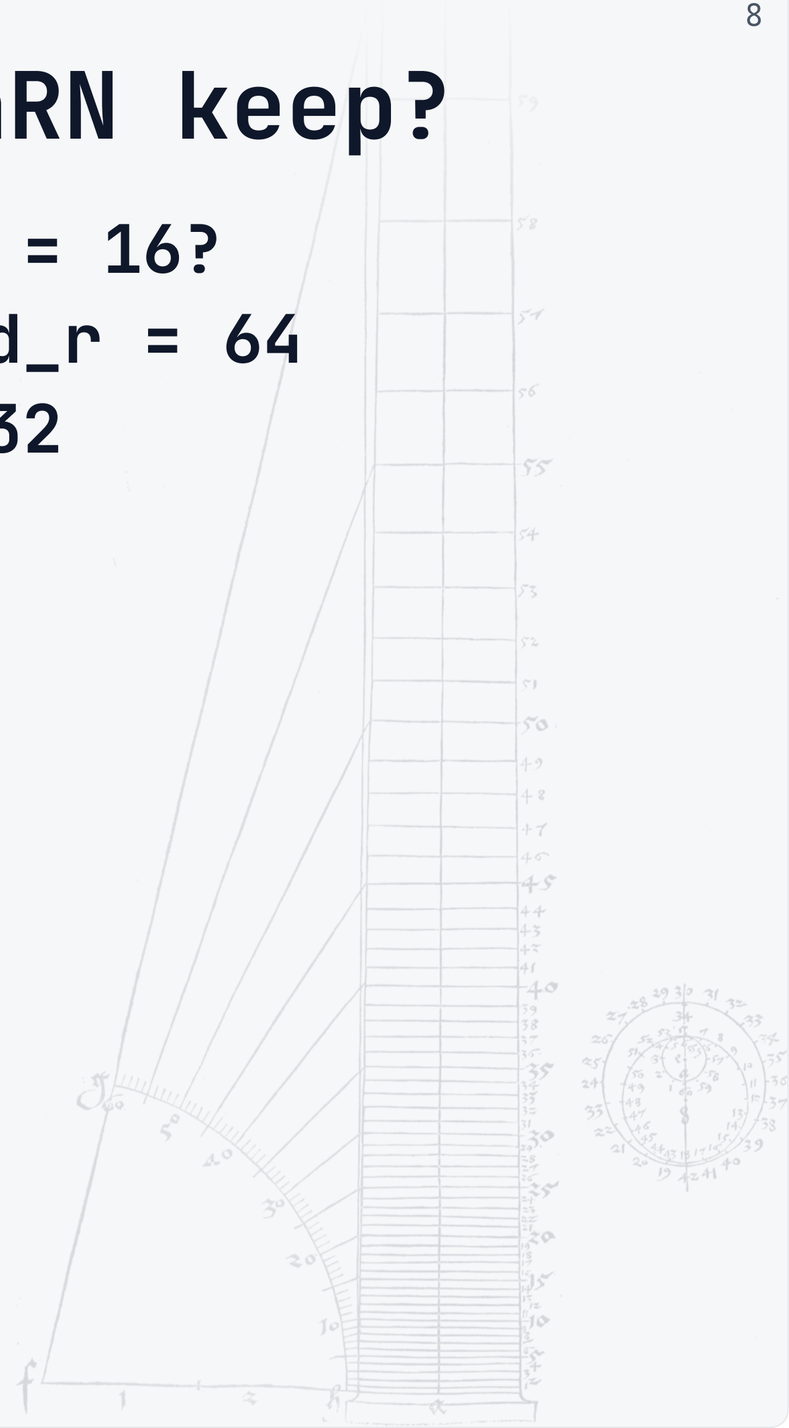
- **YaRN** (Peng et al., 2024)
- Few rotations: interpolate like PI
- Many rotations: keep the original frequency
- In between: blend, called NTK-by-parts

# Which pairs does YaRN keep?

What is γ for pair j = 16?

L_0 = 8K, b = 10,000, d_r = 64

ramp α = 1, β = 32



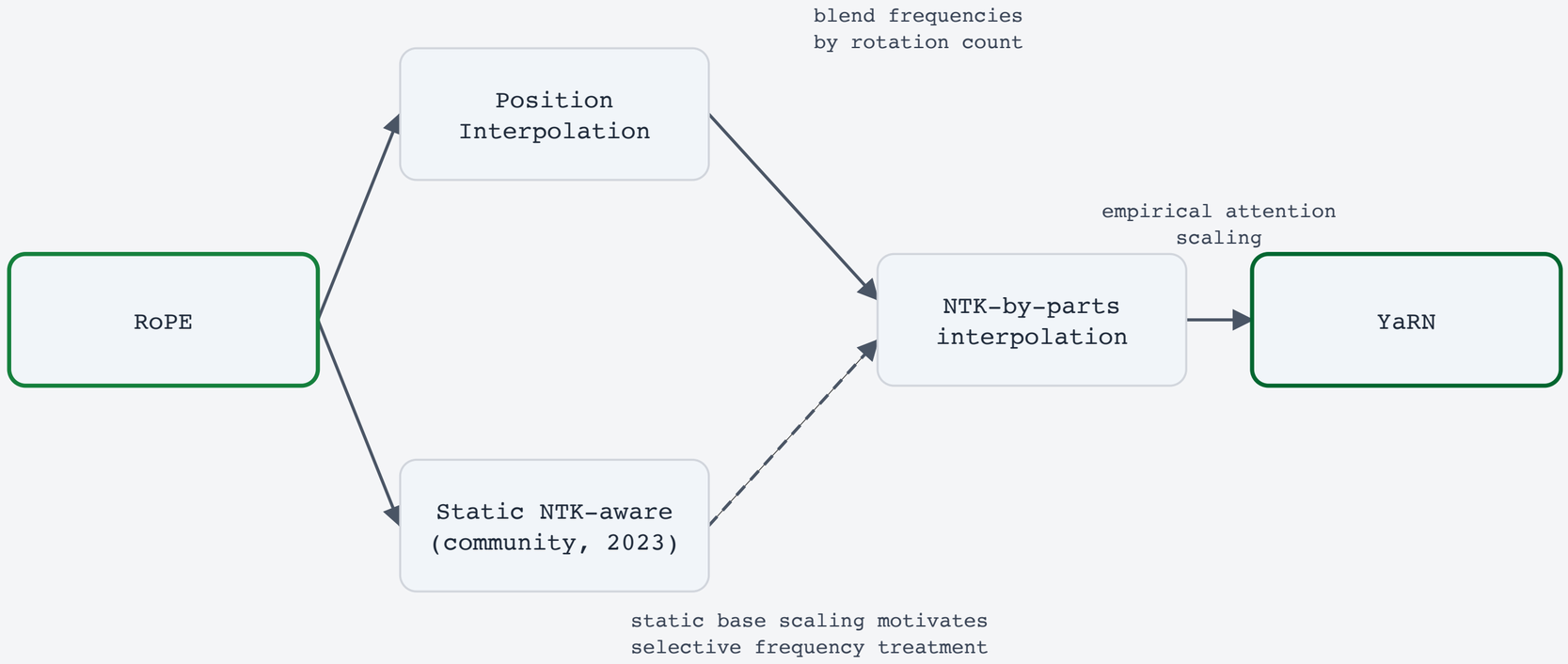
# At 8K, YaRN keeps 13 pairs and interpolates 7

- Pairs 0–12: over 32 turns, kept
- Pairs 25–31: under one turn, interpolated
- Pairs 13–24: blended
- Pair $j = 16$: $r = 13.04$ and $\gamma = 0.388$

# YaRN adds attention scaling to the blend

$$a(\rho) = 0.1 \ln \rho + 1 \tag{6}$$

- Factor $a$ fitted per ratio on Llama models (Peng et al., 2024)
- YaRN multiplies queries and keys by $a$
- Keeps attention entropy from growing with more keys



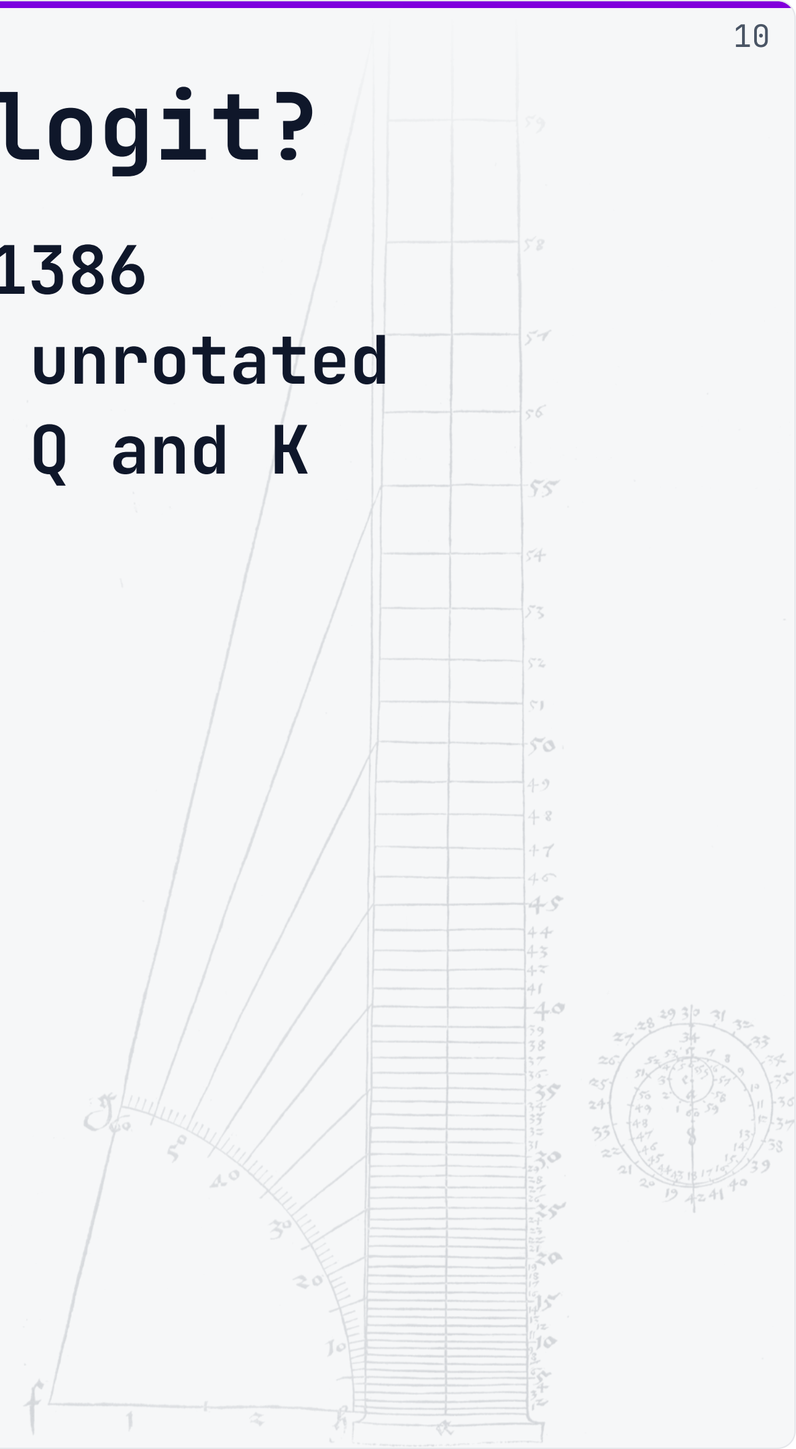

Schematic after Peng et al. (ICLR 2024), Figure 1.

# What is the new logit?

$\rho = 4$ gives $a \approx 1.1386$

$\ell_r = 2$ rotated, $\ell_u = 3$ unrotated

scale only the rotated Q and K



# Scaling only rotated coordinates gives 5.59

$$a^2 \ell_r + \ell_u \approx 1.2965 \cdot 2 + 3 \approx 5.593$$

- Scaling the whole logit instead gives 6.482
- DeepSeek-V3's code scales the whole logit (DeepSeek-AI, 2025)

# LongRoPE searches the rescale factors

- **LongRoPE** (Ding et al., 2024)
- Searched: one rescale factor per pair
- Searched: where rescaling starts
- Fine-tune to 256K, then search to 2,048K
- Factors readjusted for short contexts
- Used by the 128K Phi models (Abdin et al., 2024) (Microsoft et al., 2025)

# Three interventions, three costs

| Intervention | Question it answers | Main cost |
| --- | --- | --- |
| Inference-only remapping | Serve longer without training? | None up front; must pass the evaluation matrix |
| Continued long-context training | Can you afford to adapt the weights? | Training compute and long data |
| Target-length training | Must weights see the full length? | Attention, memory, batch and data |

# Real recipes mix training and remapping

| Model | Training for length | Remapping |
|---|---|---|
| Llama 3 (Grattafiori et al., 2024) | Base 500,000 from the start; 405B: six stages, 8K to 128K | None in the report; Llama 3.1 code rescales by parts (Meta, 2025) |
| Qwen3 (Yang et al., 2025) | ABF to 1,000,000; a 32K stage | YaRN and Dual Chunk Attention (DCA) at inference |
| OLMo 3 (Ettinger et al., 2025) | 8K to 65K on 50B to 100B tokens | YaRN on full-attention layers only |
| DeepSeek-V3 (DeepSeek-AI et al., 2024) | Two 1,000-step phases, 4K to 32K to 128K | YaRN on the RoPE part only |
| Long-context Phi (Abdin et al., 2024) | No length stage reported | LongRoPE, 4K to 128K |

# Remapping pulls angles back; training improves use

- Remapping pulls angles back toward the trained range
- Remapping alone teaches nothing about distant evidence
- More extension tokens improved OLMo 3's long-context scores (Ettinger et al., 2025)
- PI's authors fine-tuned briefly after remapping (Chen et al., 2023)
- Longer sequences: more attention work and memory

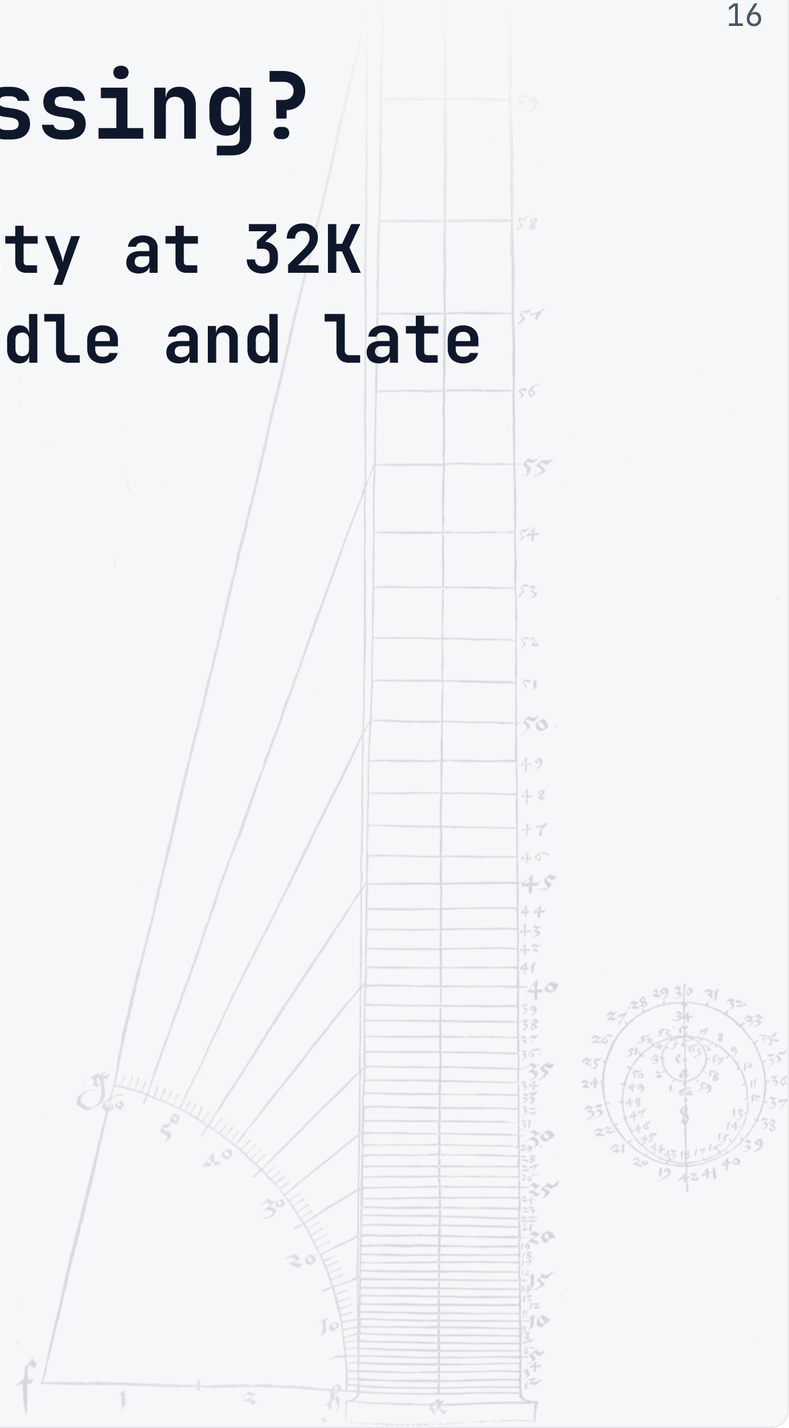
# Context length is not context use

| Check | Concrete 8K to 32K design |
| --- | --- |
| Language modeling | Negative log-likelihood (NLL) at 8K, 16K, 24K, 32K |
| Retrieval | Evidence at early, middle and late positions |
| Distant-evidence use | Question answering (QA), aggregation or summaries needing that evidence |
| Original window and serving | 8K loss and tasks, short prompts, latency, memory |
| Robustness | Several prompts and documents; seeds or intervals |
| Reproducibility | Tokenizer, effective target, prompt format, decoding |
| Decision | Task floor and allowed regression, set in advance |

# What is still missing?

## 8K to 32K: flat perplexity at 32K

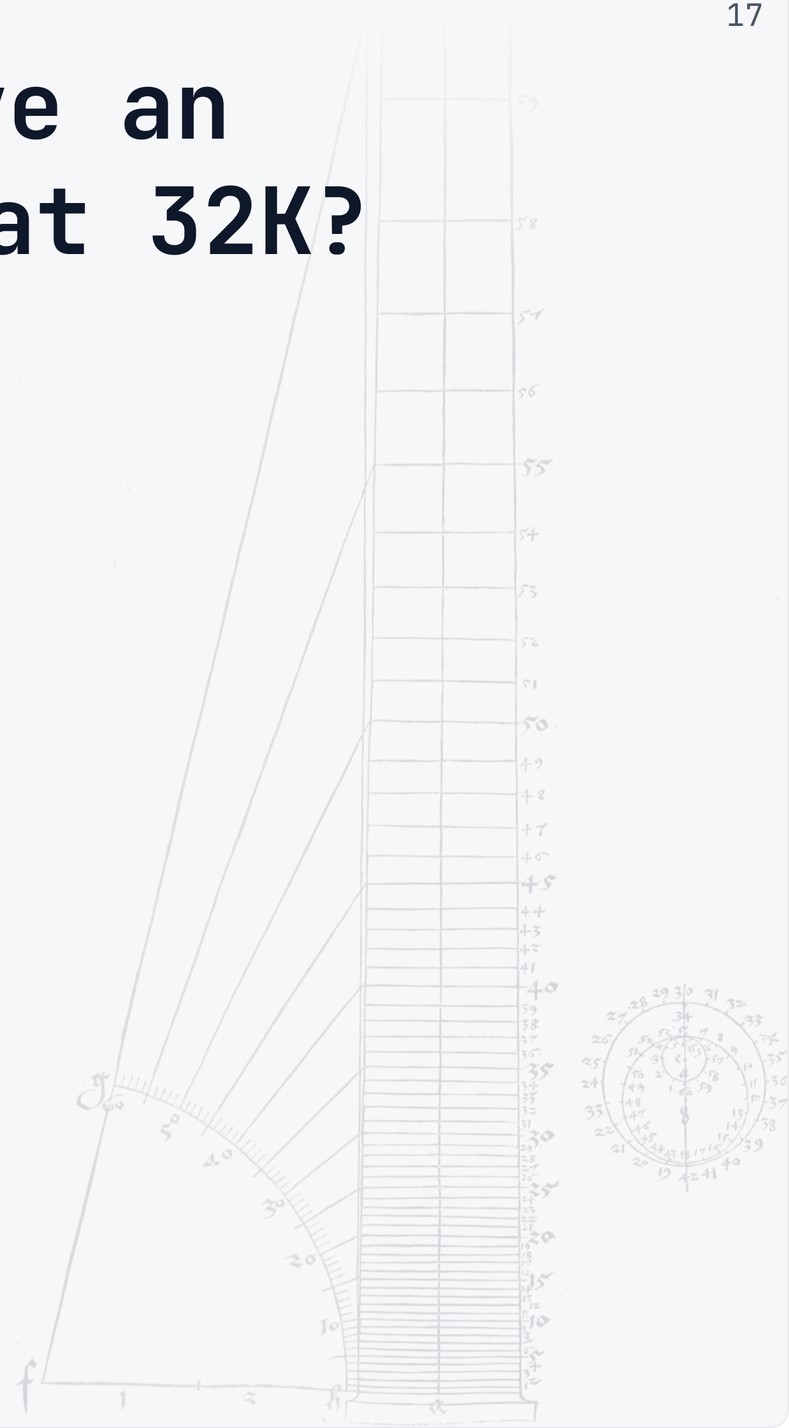
## retrieval passes early, middle and late



# Flat perplexity and retrieval are not enough

- A task that needs distant evidence
- Quality at 8K, plus serving latency and memory
- Several prompts, documents and seeds
- Tokenizer and decoding details recorded
- A threshold set before the run

# So why not serve an 8K-trained model at 32K?



# Remap the angles, train for use, measure both

- Every position still gets an angle
- Slow pairs leave the trained range
- Remapping pulls the angles back toward the trained range
- Training on long data improves use of distant tokens
- Only the evaluation matrix decides

# Six results on long context

- Past $L_0$, slow pairs meet unseen angles
- PI scales all pairs, while NTK spares the fastest
- YaRN: blend by rotation count, scale attention
- LongRoPE: searched per-pair factors
- Recipes mix training and remapping
- Test context use, not context length

# Reflection questions

No submission. Bring short answers to the Week 4 tutorial.

1. Take $b = 10{,}000$, $d_r = 64$ and an 8K-trained model served at 32K, so $\rho = 4$. Under static NTK-aware scaling, which of the seven pairs that never complete a turn within 8K still go beyond their largest trained angle at 32K, and by what factor does each exceed it? What do PI and YaRN do to the same seven pairs?
2. Your course model uses full-head RoPE with $b = 10{,}000$ and $D_h = 128$, trained on 1,024-token windows. It must now answer questions about 4,096-token documents, and you can afford one short continued-training run. Choose how to change the rotary frequencies and with which settings, say what the run trains on, and state the evaluation and the acceptance threshold you would fix before the run.
3. An 8K-trained model with full rotary, $b = 10{,}000$ and $d_r = 64$ must serve inputs of up to 32K tokens without any training, but most of its prompts are under 2K tokens. Compare PI and YaRN, both at $\rho = 4$: which pairs and which logits of a 1K prompt does each change, and by how much? Which would you deploy, and what must you measure before you decide?